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Faculty of Medicine and Health Sciences

Division of Epidemiology and Biostatistics

Title: *Design and report simulation studies the smart way with ADEMP*

Abstract

Simulation studies are the unsung heroes of modern biostatistics, the silent engines that power our understanding of how statistical methods behave under realistic conditions. Yet, many are poorly designed or unclearly reported, leaving results open to misinterpretation.

Enter ADEMP; a structured, five-step framework that transforms how we plan, execute, and report simulation studies. Standing for Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures, ADEMP provides a simple yet powerful roadmap to ensure transparency, reproducibility, and scientific rigour.

In this presentation, we'll unpack each component of ADEMP with real-world examples, explore common pitfalls (and how to avoid them), and showcase a worked simulation example inspired by Morris et al. (2019, *Statistics in Medicine*). Whether you're new to simulation or looking to sharpen your reporting, you'll leave with practical tools to design smarter, more credible simulation studies.

About the Presenter



Alexander van Twisk is a master's student in Biostatistics at Stellenbosch University. He holds a BSc in Applied Mathematics, a Postgraduate Certificate in Education, and a BSc Honours in Biostatistics. His academic path reflects a passion for statistical reasoning, clear communication, and modern research practices. Alexander is particularly interested in statistical programming in R and the safe, efficient use of AI to enhance biostatistical analysis and reproducibility.

When: 17 October, 12:00

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